

Ojas Srivastava

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Software Engineer Intern — Monomer Research

B.Tech Artificial Intelligence, SVNIT Surat | **Graduating Batch 2028** (Pre-final Year) | CGPA **9.20/10**

Modern C++ | **Linux** | **Data Structures & Algorithms** | **Operating Systems** | **Computer Architecture**

Summary

Pre-final-year **B.Tech** student (Batch **2028**), CGPA **9.20/10**, seeking a **Software Engineer Intern** role building and optimizing **high-performance** systems in **modern C++** on **Linux**. Primary implementation language for competitive programming and systems tooling is **C++ (C++17, STL, -O2)**. Strong foundation in **Data Structures & Algorithms, Operating Systems, and Computer Architecture**; experienced profiling and tuning latency-sensitive paths and concurrent services. No prior trading background required — motivated to apply systems and algorithmic skills to high-performance trading software.

Education

Bachelor of Technology in Artificial Intelligence (CS-related)

Sardar Vallabhbhai National Institute of Technology (SVNIT), Surat, India

CGPA: 9.20/10

Aug 2024 – May 2028

- **Eligibility:** Graduating batch of **2028** (pre-final year); CGPA **9.20/10** (above 7.0); no active backlogs
- **Relevant Coursework:** **Data Structures & Algorithms, Operating Systems, Computer Organization & Architecture, Object-Oriented Programming (C++), Computer Networks, DBMS, Distributed Systems, Probability & Statistics**

Competitive Programming (Primary Language: Modern C++)

- **LeetCode Knight:** Max rating **2048, 711+** problems, 32 contests — intensive **DSA** practice in **C++** (leetcode.com/Oju_Srivastava)
- **Codeforces Specialist:** Max rating **1419, 252+** solved — contest implementation under time pressure in **C++** (codeforces.com/profile/Oju)
- Daily focus on complexity analysis, memory/time trade-offs, and writing correct, fast **C++** under constraints

Projects (C++ / Linux / Performance)

OA Forge — C++ Online Assessment Judge (Linux Toolchain)

Jun 2026 –

Present

GitHub

Modern C++17, Linux, g++, STL, subprocess sandboxing, PostgreSQL

- Built a hybrid **C++** code judge on **Linux**: local **g++** compilation (`-std=c++17 -O2`), subprocess sandboxing, timed execution, and automated verdicts across **14K+** question occurrences and **5.5K+** test cases.
- Designed **object-oriented** harness generation and STL-based I/O parsing — systems-style tooling close to how high-performance compilers/runners exercise code under resource limits.

LogiFlow — Latency-Aware Large-Scale Pipeline (Google Solution Challenge 2026, Global Top 106)

Apr 2026 – Jun

2026

GitHub | Live

C++, Python, Redis, SQL, FastAPI, GCP Cloud Run, pytest

- Engineered **high-performance** read paths (**100–400 ms**) over **580+ corridors** / **9,526 stations** by separating hot-path caching (Redis) from heavy compute — optimize-for-latency mindset.
- Used **C++/Python** where each fit; owned reproducible validation (**100/100** pytest rules) so performance changes stay correct under load-like data volumes (**15,650** train-day records).

AirHelp — Concurrent Real-Time Services (PowerMind Hackathon 2026)

2026

Python, FastAPI, WebSockets, graph algorithms (A*), Linux deploy

GitHub

- Built concurrent real-time services; profiled and tuned end-to-end latency to **1–3s** for **50–100** concurrent users — applied algorithm design (A*) with complexity-aware implementation.

Experience

Software Engineering Intern

IFFCO (Indian Farmers Fertiliser Cooperative), Phulpur Unit, Prayagraj, India

Jun 2025 – Jul

2025

- Designed, developed, tested, and deployed production services on **Linux** (Docker, CI/CD): **10+ REST APIs**, optimized SQL, and debugging of production failures under tight timelines.
- Owned reliability end-to-end — unit/integration tests, code review, and clear communication of technical trade-offs.

Technical Skills

Modern C++ on Linux: C++17, STL, g++, -O2, object-oriented design, debugging, Git; comfortable developing and running tooling on **Linux**

Core CS (role-critical): Data Structures & Algorithms, Operating Systems, Computer Organization & Architecture, complexity analysis, concurrency basics

Performance & systems: Latency optimization, caching, concurrent/real-time services, profiling mindset, network programming (TCP/IP, WebSockets)

Also: Python, C, SQL, Redis, Docker, CI/CD, pytest

Leadership

- **Google Solution Challenge 2026 — Global Top 106; Vibe2Ship 2026 — Global Top 20**
- **Executive Member, ACM SVNIT & Mentor, Nexus SVNIT** — DSA workshops and contest mentoring in **C++**